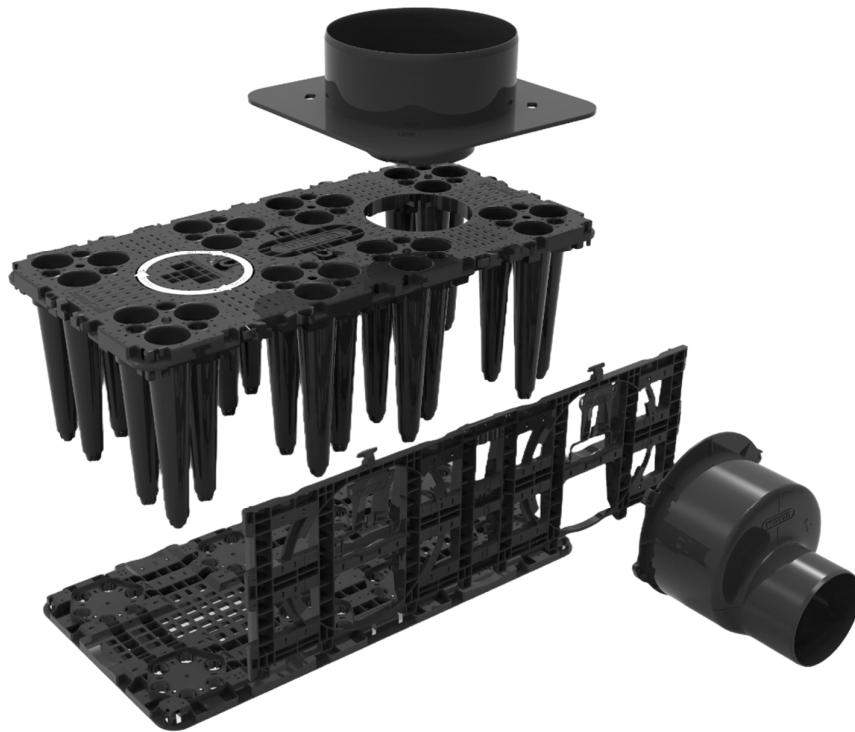


November 11, 2024

RE: Vehicle loading over Wavin AquaCell stormwater units in Standard Configuration (SD)

To whom it may concern:

This letter is in response to a request for specific information on the load bearing capability of the Wavin AquaCell in its SD configuration. This configuration of the AquaCell consists of a base unit module connected to a bottom plate. Side plates are added as needed to create a fully closed module. Additional modules are connected via integrated clips on the top plate of the base unit. The seamless connection of the AquaCell units, with no additional clips or connectors needed, will form a complete, highly configurable multi-layer tank specific to the design and application of the end user.



Independent testing on the AquaCell was conducted by TRI Environmental in Austin, TX, in April of 2024 (Reference TRI log number 24-001165). Compression testing was based on a bench-scale compression test as referenced in ASTM D2412. Compressive load was applied to the modules as directed in ASTM F2418 and ASTM F2922. Load was applied until module failure. *No backfill or finished pavement was present during testing (test module only).*

The average maximum compressive load was reported at 77,779 lbs., with maximum compressive stress reported at 70.0 psi.

Please reference Wavin documents TB101 & TRI log 24-001165 for further information.

Wavin as been requested to analyze the vehicle axle loads provided below, as supported by an asphalt concrete paving section, and bearing over an AquaCell SD system. In considering the performance of the AquaCell given the provided axle loads, additional load bearing reinforcement is provided by a layer of biaxial geogrid installed 12 inches above the top of the AquaCell tank per the manufacturer recommendations.

Stormwater structural loading design vehicles provided:

- AquaCell SD Units: Maximum cover to top of AquaCell SD Units = 18 inches
- Vehicle Axle Load, HS20 = Maximum Dual Tire Rear Axle Load of 32,000 pounds
- Standard Vactor Truck Axle Load = Maximum Dual Tire Rear Axle Load of 33,665 pounds
- Large Vactor Truck Axle Load = Maximum Single Tire Rear Axle Load of 22,875 pounds
- Large Vactor Truck Axle Load = Maximum Dual Tire Rear Axle Load of 22,875 pounds
- Standard Fire Truck Axle Load = Maximum Dual Tire Rear Axle Load of 24,000 pounds

Analysis

A. AquaCell SD Units, Traveling Vehicle HS20 Axle Loading - 18 Inch Cover

- HS20 Load: 32,000 pounds applied to Dual Tire Axle (16,000 per Dual Tires)
- Surface Contact Area Loaded Dual Tires: 20 x 10 inches = 200 Squares Inches
- Dynamic (Moving Vehicle) Loading Factor = 1.3
- Vertical Compressive (Crush) Strength of AquaCell SD Unit = 70.0 psi
- Distance from top of installed AquaCell Tanks to surface = 18 inches
- Assumed unit weight of fill over the AquaCell Tanks = 130 pcf
- Simplified applied surface load distribution with depth: area of load application increases based on a 2 vertical:1 slope

Projected Tire Load Area at top of AquaCell: $(20 + 18) \times (10 + 18) = 1,064$ sq. in.

Projected Tire Load at top of AquaCell: $[(16,000) / 1,064] \times 1.3 = 19.54$ psi

Vertical fill load at the top of AquaCell: $(18/12) \times (130 \text{ pcf}) / 144 = 1.35$ psi

Total vertical Load at top of AquaCell: $19.54 + 1.35 = 20.89$ psi

Factor of Safety (FS) to AquaCell SD Crushing: $70.0 / 20.89 = 3.35$ (Simplified to FS=3.0)

B. AquaCell SD Units, Standard Vactor Truck Dual Tire Axle Load - 18 Inch Cover

- Load: 33,665 pounds applied to Dual Tire Axle (16,832 per Dual Tires)

- Surface Contact Area Loaded Dual Tires: 20 x 10 inches = 200 Squares Inches
- Dynamic (Moving Vehicle) Loading Factor = 1.3
- Vertical Compressive (Crush) Strength of AquaCell SD Unit = 70.0 psi
- Distance from top of installed AquaCell to surface = 18 inches
- Assumed unit weight of fill over the AquaCell tank = 130 pcf
- Simplified applied surface load distribution with depth: area of load application increases based on a 2 vertical: 1 slope

Projected Tire Load Area at top of AquaCell Tanks: $(20 + 18) \times (10 + 18) = 1,064$ sq. in.

Projected Tire Load at top of AquaCell Tanks: $[(16,832) / 1,064] \times 1.3 = 20.56$ psi

Vertical fill load at the top of AquaCell Tanks: $(18/12) \times (130 \text{ pcf}) / 144 = 1.35$ psi

Total vertical Load at top of AquaCell Installation: $20.56 + 1.35 = 21.91$ psi

Factor of Safety (FS) to AquaCell SD Tank Crushing: $70.0/21.91 = 3.19$ (Simplified to FS=3)

C. AquaCell SD Units, Large Vector Truck Single Tire Axle Load - 18 Inch Cover

- Load: 22,875 pounds applied to Single Tire Axle (11,473 per Tire)
- Surface Contact Area Loaded Dual Tires: 10 x 10 inches = 100 Squares Inches
- Dynamic (Moving Vehicle) Loading Factor = 1.3
- Vertical Compressive (Crush) Strength of AquaCell SD Unit = 70.0 psi
- Distance from top of installed AquaCell Tanks to surface = 18 inches
- Assumed unit weight of fill over the AquaCell Tanks = 130 pcf
- Simplified applied surface load distribution with depth: area of load application increases based on a 2 vertical:1 slope

Projected Tire Load Area at top of AquaCell: $(10 + 18) \times (10 + 18) = 784$ sq. in.

Projected Tire Load at top of AquaCell Tank: $[(11,473) / 784] \times 1.3 = 19.02$ psi

Vertical fill load at the top of AquaCell Tank: $(18/12) \times (130 \text{ pcf}) / 144 = 1.35$ psi

Total vertical Load at top of AquaCell Tank Installation: $19.02 + 1.35 = 20.52$ psi

Factor of Safety (FS) to AquaCell SD Crushing: $70.0/20.52 = 3.41$ (Simplified to FS=3.4)

D. AquaCell SD Units, Large Vector Truck Dual Tire Axle Load - 18 Inch Cover



Building &
Infrastructure

- Load: 22,875 pounds applied to Dual Tire Axle (11,437 per Dual Tires)
- Surface Contact Area Loaded Dual Tires: 20 x 10 inches = 200 Squares Inches
- Dynamic (Moving Vehicle) Loading Factor = 1.3
- Vertical Compressive (Crush) Strength of ACF AquaCell SD Unit = 70.0 psi
- Distance from top of installed AquaCell Tanks to surface = 18 inches
- Assumed unit weight of fill over AquaCell Tanks = 130 pcf
- Simplified applied surface load distribution with depth: area of load application increases based on a 2 vertical:1 slope

Projected Tire Load Area at top of AquaCell Tanks: $(20 + 18) \times (10 + 18) = 1,064$ sq. in.

Projected Tire Load at top of AquaCell Tanks: $[(11,437) / 1,064] \times 1.3 = 13.97$ psi

Vertical fill load at the top of AquaCell Tanks: $(18/12) \times (130 \text{ pcf}) / 144 = 1.35$ psi

Total vertical Load at top of AquaCell Tank Installation: $13.97 + 1.35 = 15.32$ psi

Factor of Safety (FS) to AquaCell SD Crushing: $70.0/15.32 = 4.57$ (Simplified to FS=4.6)

E. AquaCell SD Units, Fire Truck Dual Tire Axle Load - 18 Inch Cover

- Load: 24,000 pounds applied to Dual Tire Axle (12,000 per Dual Tires)
- Surface Contact Area Loaded Dual Tires: 20 x 10 inches = 200 Squares Inches
- Dynamic (Moving Vehicle) Loading Factor = 1.3
- Vertical Compressive (Crush) Strength of AquaCell Tank SD Unit = 70.0 psi
- Distance from top of installed AquaCell Tanks to surface = 18 inches
- Assumed unit weight of fill over the AquaCell Tanks = 130 pcf
- Simplified applied surface load distribution with depth: area of load application increases based on a 2 vertical:1 slope

Projected Tire Load Area at top of AquaCell Tanks: $(20 + 18) \times (10 + 18) = 1,064$ sq. in.

Projected Tire Load at top of AquaCell Tanks: $[(12,000) / 1,064] \times 1.3 = 14.66$ psi

Vertical fill load at the top of AquaCell Tanks: $(18/12) \times (130 \text{ pcf}) / 144 = 1.35$ psi

Total vertical Load at top of AquaCell Tank Installation: $14.66 + 1.35 = 16.01$ psi

Factor of Safety (FS) to AquaCell SD Tank Crushing: $70.0/16.01 = 4.37$ (Simplified to FS=4.4)

Information presented in this letter has been prepared utilizing information provided to Wavin and in accordance with generally accepted engineering practices.

Additionally, this letter is based on the assumption that all Wavin AquaCell units will be installed per guidelines provided by the manufacturer. Any damage to the AquaCell during the installation of the stormwater system (configuration, connection of units, placement of backfill, and construction of the surface pavement) may impact the load bearing performance of the AquaCell in its intended application.

If any additional information is needed, please contact Wavin directly at wavintechsupport@orbia.com.

Regards,

James Duggan

Technical Support Manager

Wavin, North America

Heath.duggan@orbia.com

Attachments: TB101-AquaCell Load Testing Technical Bulletin; 24-0011165-Wavin-Stormwater-Chamber-Compression-AquaCell Standard-Report

AquaCell

Technical Bulletin TB101

AquaCell load capacity testing results by TRI Environmental

Objective

1. Determine the load carrying capacity of the AquaCell system in a controlled environment.

Test Method

2. AquaCell units (standard and extra strong configurations) were subjected to a compressive load in a controlled environment until failure.
 - a. Compression testing was based on a bench-scale compression test ASTM D2412.
 - b. Test was modified to apply compressive load at constant 2.0% rate at height of chamber per minute as directed by ASTM F2418 & ASTM F2922.
 - c. Load was applied until module failure, or the maximum capacity of 135,000 lbs. was reached.
 - d. No backfill or finished pavement was present during testing (test module only).

Test Results

3. Standard Configuration:
 - a. Average maximum compressive load: 77,779 lbs.
 - b. Exceeds HS-20 load rating (72,000 lbs)

Product	Test Specimen	Maximum Compressive Load (lb)	Maximum Compressive Stress (psi)	Extension at Max Load (in)
AquaCell Standard Configuration	1	76,824	69.2	0.926
	2	78,043	70.3	0.943
	3	78,470	70.7	0.949
	Avg.	77,779	70.0	0.939

4. Extra Strong Configuration:
 - a. Average maximum compressive load: 117,839 lbs.
 - b. Exceeds HS-25 load rating (90,000 lbs)

Product	Test Specimen	Maximum Compressive Load (lb)	Maximum Compressive Stress (psi)	Extension at Max Load (in)
AquaCell Extra Strong Configuration	1	117,681	106.0	1.143
	2	118,891	107.1	1.155
	3	116,944	105.3	1.152
	Avg.	117,839	106.1	1.150

Discussion

5. The AquaCell system demonstrated significant load carrying capacity in both standard and extra strong configurations. The standard configuration achieved an average maximum compressive load exceeding the HS-20 rating, with the expectation that backfill, and finished pavement will further improve load distribution and performance. The extra strong configuration exceeded the HS-25 load rating, suitable for the most demanding commercial applications.

Conclusion

6. Based on testing, the AquaCell system can be utilized in the following load classifications
 - a. ASHTO H-20, HS-20, H-25 (Standard Configuration)
 - b. AASHTO HS-25 (Extra Strong Configuration)

Note

1. AASHTO refers to the American Association of State Highway and Transportation Officials.
2. Testing completed at TRI Environmental, Austin, Texas.
 - a. Results referenced from TRI log number 24-001165 dated April 23, 2024.

Disclaimer

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TESTING, RESEARCH, CONSULTING AND FIELD SERVICES

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April 23, 2024

Mr. Kraig Smith
Orbia Building & Infrastructure (Wavin)
950 Winter Street, South Entrance 1st Floor
Waltham, MA 02451
email: kraig.smith@orbia.com

Re: Stormwater Chamber Compression Test Report

Dear Mr. Smith:

TRI/Environmental, Inc. (TRI) is very pleased to present this final test report for laboratory testing of Aquacell Stormwater Chambers.

Testing

TRI received samples of Aquacell underground stormwater storage chambers for testing and analysis. The samples were received and tested under TRI log number 24-001165.

Compression testing was based on a bench-scale compression test, ASTM D2412, *Standard Test Method of the Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading*. The test method was modified to apply the compressive load at a constant rate of 2.0% of the height of the chamber per minute as directed in ASTM F2418, *Standard Specification for Polypropylene (PP) Corrugated Wall Stormwater Collection Chambers* and ASTM F2922, *Standard Specification for Polyethylene (PE) Corrugated Wall Stormwater Collection Chambers*. Compression testing was conducted on an Instron Model 5889 Tensile/Compression Testing Machine equipped with a 135k pound capacity load cell and Blue Hill data acquisition software. Reinforced loading platens were used for applying the compressive load. Load was applied until specimen failure or the maximum capacity of 135k pounds was reached.

Results

Product	Test Specimen	Maximum Compressive Load(lb)	Maximum Compressive Stress (psi)	Extension at Max Load (in)
Aquacell Standard Configuration	1	76,824	69.2	0.926
	2	78,043	70.3	0.943
	3	78,470	70.7	0.949
	Avg.	77,779	70.0	0.939

Photos



Pre-Test



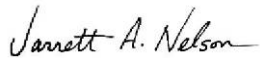
Post-Test



Post-Test

TRI is very pleased to present this final report for testing services. If you have any questions or require any additional information, please contact me at JNelson@tri-env.com or telephone number 512-263-5944.

Sincerely,



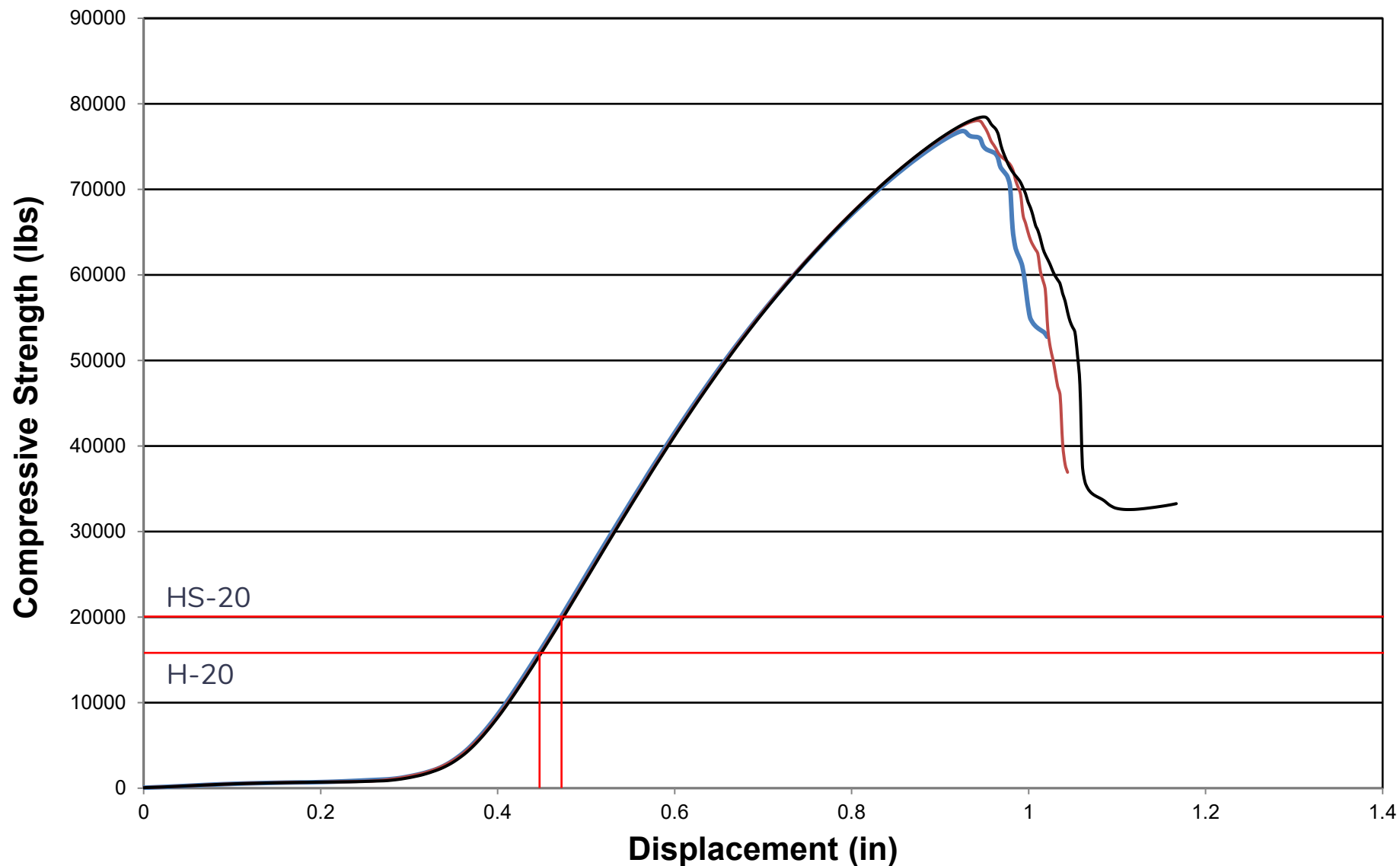
Jarrett A. Nelson
Director of Operations
TRI Geosynthetic Services
www.GeosyntheticTesting.com

Attachments: Stress/Strain Curves



Compression Test Results

TRI Client: Wavin USA
Product: Aquacell Standard





Compression Test Results
TRI Client: Wavin USA
Product: Aquacell Standard

